

Module 4

Continuous-Time Fourier Transform (CTFT)

Section A — Basic Problems

Problem 1

Generate the continuous-time rectangular pulse

$$x(t) = \begin{cases} 1, & |t| \leq 0 \\ 0, & \text{otherwise} \end{cases}$$

Plot the signal and identify width, amplitude and energy.

Problem 2

Generate the sinc function

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

Observe:

- Main lobe
- Side lobes
- Symmetry

Problem 3

Numerically compute and plot the Fourier Transform of a rectangular pulse.

Problem 4

Plot the magnitude spectrum of a sinc function.

Problem 5

Compare:

- Narrow rectangular pulse

- Wide rectangular pulse

How does pulse width affect the spectrum?

Section B — Intermediate Problems

Problem 6

Verify the Time Shifting Property of the Fourier Transform.

Problem 7

Verify the Frequency Shifting Property.

Problem 8

Verify the Scaling Property.

Problem 9

Verify the Duality Property.

Problem 10

Verify the Differentiation Property.

Section C — Programming Problems

Problem 11

Numerically compute the CTFT of

$$x(t) = e^{-t}u(t)$$

Problem 12

Compare the spectra of:

- Rectangular pulse

- Triangular pulse
- Gaussian pulse

Problem 13

Investigate how increasing signal duration changes bandwidth.

Problem 14

Demonstrate the Convolution Theorem numerically.

Problem 15

Verify Parseval's Relation for the Fourier Transform.

Challenge Problems

Challenge 1

Design a simple low-pass filter in the frequency domain.

Challenge 2

Remove high-frequency noise from a signal using FFT.

Challenge 3

Compare time-domain multiplication with frequency-domain convolution.

Challenge 4

Generate a chirp signal and analyze its spectrum.

Challenge 5

Compare FFT results for different sampling frequencies.

Challenge 6

Build a mini "Spectrum Analyzer" that plots:

- Time-domain signal
- Magnitude spectrum
- Phase spectrum

for any user-defined signal.